Target validation paths — demo-nsclc-egfr-l858r-post-osi-d3m0

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the savolitinib + osimertinib trial enrollment (SAFFRON / SACHI). The validation work here is about resistance characterization at progression, not target confirmation. The goal is to make sure the case really is MET-amplified bypass resistance and not something more complicated wearing the same imaging signature.

EGFR L858R

The most consequential miss right now would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS including TP53 and RB1 status is the screen: co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold (≥ 6) cleanly, but MET amplification can be focal, and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or a matched archival block) refines confidence that the dominant disease, not just the biopsied site, is the amplified one. MET IHC (clone SP44) is the orthogonal-modality complement to the SAFFRON eligibility profile. A comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

Where to order these assays

The preferred provider for each assay is marked **(preferred)**, selected on company size, reputation, US-based location, and turnaround time. Other providers in the row are listed in case the preferred lab is unreachable for this patient.

Assay	Provider	Decision gated	Contact
ctDNA panel (EGFR C797S / T790M / exon-20)	**Guardant Health (preferred) (Guardant360 CDx)**	**Distinguishes on-target (C797S, exon-20) vs bypass post-osimertinib resistance; informs fourth-gen TKI vs combination strategy**	**test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887**
ctDNA panel	Foundation Medicine (FoundationOne Liquid CDx)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
ctDNA panel	Tempus (xF+)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137

ctDNA panel	Caris Life Sciences (Caris Assure)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
ctDNA panel	NeoGenomics Laboratories (NeoLAB Liquid)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor NGS for TP53 / RB1 + bypass alterations	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Predicts SCLC histologic transformation; informs whether next-line stays adenocarcinoma-targeted or pivots to chemo**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Tumor NGS	Caris Life Sciences (Molecular Intelligence)	Predicts SCLC histologic transformation	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS	Tempus (xT)	Predicts SCLC histologic transformation	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive)	Predicts SCLC histologic transformation	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor NGS	Memorial Sloan Kettering (MSK-IMPACT)	Predicts SCLC histologic transformation	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Re-biopsy + neuroendocrine IHC panel	**Mayo Clinic Laboratories (preferred)**	**Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected; pivots therapy entirely**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
Re-biopsy + neuroendocrine IHC	ARUP Laboratories	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
Re-biopsy + neuroendocrine IHC	LabCorp / Esoterix Oncology	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
Re-biopsy + neuroendocrine IHC	Quest Diagnostics	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Re-biopsy + neuroendocrine IHC	Memorial Sloan Kettering Diagnostic Molecular Pathology	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
MET FISH (gene/CEP7 ratio)	**NeoGenomics Laboratories (preferred)**	**Refines confidence in MET amp gating result for SAFFRON enrollment; does not gate eligibility**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
MET FISH	Foundation Medicine	Refines MET amp gating result; does not gate eligibility	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639

MET FISH	Mayo Clinic Laboratories	Refines MET amp gating result; does not gate eligibility	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
MET FISH	LabCorp / Esoterix Oncology	Refines MET amp gating result; does not gate eligibility	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MET FISH	Quest Diagnostics	Refines MET amp gating result; does not gate eligibility	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MET IHC (clone SP44)	**Mayo Clinic Laboratories (preferred)**	**Confirms MET protein-level expression; orthogonal complement to FISH but does not change SAFFRON eligibility**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
MET IHC	ARUP Laboratories	Confirms MET protein-level expression	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
MET IHC	NeoGenomics Laboratories	Confirms MET protein-level expression	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MET IHC	LabCorp / Esoterix Oncology	Confirms MET protein-level expression	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MET IHC	Quest Diagnostics	Confirms MET protein-level expression	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Comprehensive bypass NGS (HER2 / BRAF / KRAS / FGFR)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Frames durability expectations and the next-line plan for MET-directed therapy; does not gate SAFFRON**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Comprehensive bypass NGS	Caris Life Sciences (Molecular Intelligence)	Frames durability and next-line plan for MET-directed therapy	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive bypass NGS	Tempus (xT)	Frames durability and next-line plan for MET-directed therapy	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
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Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

ctDNA panel including EGFR C797S, T790M, and exon 20 insertions	After osimertinib progression, the resistance landscape divides cleanly between on-target EGFR mutations (C797S, exon-20 inserts, less commonly T790M) and bypass mechanisms. Detecting C797S in cis vs trans changes whether a fourth-generation EGFR-TKI or a combination strategy is the rational next move. Skipping ctDNA at progression risks treating the case as bypass-resistance-only when the biology is mixed.	Guardant Health (Guardant360 CDx) · test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887	10–20 mL whole blood; archival tissue not required
Tumor NGS including TP53 and RB1 status	TP53 + RB1 co-loss is the strongest predictor of small-cell histologic transformation as a resistance mechanism in EGFR-mutant NSCLC progressing on osimertinib. The probability of transformation is non-trivial (3–14% across cohorts), and missing it changes the next-line conversation entirely; chemotherapy backbones and platinum-etoposide enter the picture. Pair this with re-biopsy guidance below.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	archival FFPE; if exhausted, ctDNA broad panel

Re-biopsy with neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) if imaging morphology shifts	Histologic small-cell transformation is the most consequential resistance pattern that no liquid biopsy can confirm. When TP53/RB1 are co-lost or imaging shows new visceral / explosive growth atypical for adenocarcinoma, fresh tissue with a neuroendocrine IHC panel is the only way to confirm. Without this, a transformed case can be treated as adenocarcinoma indefinitely.	Mayo Clinic Laboratories · test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710	fresh biopsy of the most-active site
MET FISH on a second site (metastatic biopsy or matched archival block)	MET amplification can be focal and discordant between primary and metastatic sites in EGFR-resistant NSCLC. The patient's GCN 8.2 from a single block clears SAFFRON's threshold, but a second-site test rules out the scenario where one biopsy hits a focal amplicon and the dominant disease is unaffected. Without it, a positive trial enrollment can rest on a non-representative result.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	second-site archival FFPE if available; not gating
MET IHC (clone SP44) for orthogonal expression confirmation	FISH-confirmed MET amplification at GCN 8.2 is the load-bearing finding, and SAFFRON enrollment accepts FISH or IHC 3+. IHC adds an orthogonal modality and surfaces protein-level MET expression; concordance increases confidence that the FISH result reflects active MET signaling. Discordance (FISH-amp without IHC overexpression) is a soft signal worth flagging but does not foreclose the trial.	Mayo Clinic Laboratories · test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710	archival FFPE; same block as prior testing

Comprehensive NGS for HER2 amp, BRAF, KRAS, FGFR1-3, and bypass-pathway alterations	Bypass amplifications co-occur with MET amp in roughly 10–20% of post-osimertinib cases and modify expected response to MET-directed therapy. Detecting co-bypass alterations doesn't foreclose savolitinib + osimertinib but reframes the durability expectation and informs subsequent-line planning. Pair this with the TP53 / RB1 panel if running comprehensive NGS; same blood draw / block.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	archival FFPE; ctDNA as backup

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.